SymPy Bug Report

1 Bug 42: `solve_univariate_inequality(atan(x) < pi)` omits $x = 0$

1.1 Minimal Reproducer

```
from sympy import N, S, atan, pi, symbols
from sympy.solvers.inequalities import solve_univariate_inequality

x = symbols("x")
sol = solve_univariate_inequality(atan(x) < pi, x, relational=False)

print("solution =", sol)
print("0 in solution =", S.Zero in sol)
print("atan(0) < pi evaluates to", atan(S.Zero) < pi)
print("numeric atan(0) - pi =", N(atan(S.Zero) - pi, 50))
```

1.2 Observed Behavior

```
solution = Union(Interval.open(-oo, 0), Interval.open(0, oo))
0 in solution = False
atan(0) < pi evaluates to True
numeric atan(0) - pi = -3.1415926535897932384626433832795028841971693993751
```

SymPy returns the two punctured open intervals $(-\infty, 0) \cup (0, \infty)$, so the result explicitly omits 0. The same run evaluates the original strict inequality at 0 as true, so the omitted singleton is not a harmless endpoint convention.

1.3 Expected Behavior

The correct solution set is S.Reals, i.e. all real numbers:

$$\text{solve\_univariate\_inequality}(\arctan(x) < \pi, x, \text{relational} = \text{False}) = \mathbb{R}.$$

In particular, 0 must be included because $\arctan(0) = 0 < \pi$.

1.4 Mathematical Explanation

For real x, the principal real arctangent satisfies

$$-\frac{\pi}{2} < \arctan(x) < \frac{\pi}{2}.$$

Since $\frac{\pi}{2} < \pi$, every real value of $\arctan(x)$ is strictly less than π. Therefore the inequality $\arctan(x) < \pi$ has no real boundary point and no excluded real singleton.

The returned set $(-\infty, 0) \cup (0, \infty)$ is not equivalent to $\mathbb{R}$. It excludes $x = 0$, but direct substitution gives $\arctan(0) = 0$, and $0 < \pi$. Thus the difference is a genuine lost solution, not a different representation of the same set.

1.5 Source-Level Diagnosis

The diagnosis identifies a downstream interaction between `sympy/solvers/solveset.py` and `sympy/solvers/inequalities.py`. In the public call shown in the minimal reproducer, the inequality solver first rewrites the problem as the sign condition for $\arctan(x) - \pi$. The periodic shortcut does not apply because the expression is not periodic in x, so the solver calls `solvify` to obtain critical equality roots.

The traced call path is

$$\texttt{solve\_univariate\_inequality} \longrightarrow \texttt{solvify} \longrightarrow \texttt{solveset}.$$

The trace in the diagnosis records `periodicity(atan(x) - pi, x) -> None`, `solvify(atan(x) - pi, x, S.Reals) -> [0]`, and `continuous_domain(atan(x) - pi, x, S.Reals) -> Reals`.

The root cause appears to start in `sympy/solvers/solveset.py` in `_invert_real`, where the inverse step accepts $x = \tan(\pi) = 0$ as a solution of $\arctan(x) = \pi$ without checking the principal real range of arctan. That value is branch-invalid: $\arctan(0) = 0$, not π. The spurious equality root is then passed to `sympy/solvers/inequalities.py`. The inequality solver partitions the real line at the reported root 0, treats it as a genuine strict boundary, includes the two neighboring open intervals after sample-point testing, and excludes the critical point itself because the relation is strict. In this branch, the original inequality is not used to rescue the candidate point 0, even though direct evaluation would show that it satisfies $\arctan(x) < \pi$.

A plausible fix direction supported by the diagnosis is to reject branch-invalid arctangent inversions in `_invert_real`. A second defensive improvement would be for the inequality solver to validate candidate critical points against the original inequality before removing them solely because the relation is strict.

1.6 Additional Failing Instances

The same failure pattern appears for bounds $b > \pi/2$ where SymPy obtains a branch-invalid equality root from $\arctan(x) = b$ and then removes that point from the strict inequality $\arctan(x) < b$. For every such real bound, $\arctan(x) < b$ is true for all real x, so the expected solution is $\mathbb{R}$. The independent checks use both SymPy's symbolic truth value and an independent numerical `math.atan` comparison at the omitted point.

Input / Expression	**SymPy behavior**	**Expected behavior**
$\arctan(x) < \pi$, omitted point 0	Returned set omits 0	$\mathbb{R}$, including 0
$\arctan(x) < 3\pi/4$, omitted point -1	Returned set omits -1	$\mathbb{R}$, including -1
$\arctan(x) < 2\pi$, omitted point 0	Returned set omits 0	$\mathbb{R}$, including 0
$\arctan(x) < 5\pi/4$, omitted point 1	Returned set omits 1	$\mathbb{R}$, including 1
$\arctan(x) < 10$, omitted point $\tan(10)$	Returned set omits $\tan(10)$	$\mathbb{R}$, including $\tan(10)$
$\arctan(x) < 3\pi$, omitted point 0	Returned set omits 0	$\mathbb{R}$, including 0

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check. The closest public material found was general SymPy solver documentation and background references on the principal range of inverse trigonometric functions; no public issue, pull request, Stack Overflow report, mailing-list thread, or release note was identified for this exact inequality output or for omission of 0 from $\arctan(x) < \pi$. The deduplication verdict is a new failure mode with no public precedent. The bug remains individually actionable because it has a small reproducer, a concrete missing solution, a localized source-level mechanism, and a focused regression test.

1.8 Proposed SymPy Regression Test

```
import pytest
from sympy import S, atan, pi, symbols, tan
from sympy.solvers.inequalities import solve_univariate_inequality


@pytest.mark.parametrize("bound,point", [(pi, 0), (3*pi/4, -1), (2*pi, 0), (5*pi/4, 1),
    (10, tan(10)), (3*pi, 0)])
def test_atan_inequality_keeps_points_that_satisfy_strict_inequality(bound, point):
    x = symbols("x")
    sol = solve_univariate_inequality(atan(x) < bound, x, relational=False)
    assert S(point) in sol
    assert atan(S(point)) < bound
```